AI Customer Churn Prediction System

A Major Project Report
Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology
in
Computer Science & AI
OR Computer Science & DS

by

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It is certified that the work contained in the project report entitled "AI Customer Churn Prediction System" by the following students has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

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I/We declare that this written submission represents my/our ideas in my/our own words and where others' ideas or words have been included, I/We have adequately cited and referenced the original sources. I/We declare that I/We have properly and accurately acknowledged all sources used in the production of this report. I/We also declare that I/We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I/We understand that any violation of the above will be a cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Chapter 1

Introduction

Customer churn refers to the phenomenon where customers discontinue their subscription or stop using a company's services. In the telecom industry, churn rates typically range between 15% to 25% annually, making churn prediction a critical business problem.

Retaining existing customers is significantly less expensive than acquiring new customers. Therefore, predictive analytics techniques are widely used to identify customers who are likely to churn.

In this project, we develop a machine learning-based churn prediction system using the Telco Customer Churn dataset. Multiple classification models are implemented and compared to identify the most effective approach.

Chapter 2

Literature Review

Customer churn prediction has been extensively studied using statistical and machine learning techniques. Traditional models such as Logistic Regression provide interpretable probabilistic outputs, making them suitable baseline models.

Tree-based models such as Decision Trees and Random Forests are capable of capturing non-linear relationships between features. These models often improve predictive performance but may suffer from overfitting.

Recent advancements focus on ensemble learning and deep learning models for improved accuracy. However, interpretability remains crucial for business decision-making in churn prediction tasks.

Chapter 3

Problem Formulation and Proposed Solution

3.1 Dataset Description

The dataset contains 7043 customer records with 21 features, including:

- Demographic information (Gender, Senior Citizen, Partner, Dependents)
- Service details (Internet, Phone, Streaming, etc.)
- Account information (Tenure, Contract type, Payment method)
- Billing details (Monthly Charges, Total Charges)

The target variable is **Churn**, which indicates whether a customer has discontinued the service.

3.2 Data Preprocessing

The following preprocessing steps were performed:

- Removal of irrelevant feature (customerID)
- Conversion of TotalCharges to numeric
- Removal of records with tenure = 0
- Handling missing values
- One-hot encoding of categorical variables

- Feature scaling using StandardScaler

After preprocessing, the final dataset consisted of 7032 records and 30 features.

3.2.1 Churn Distribution

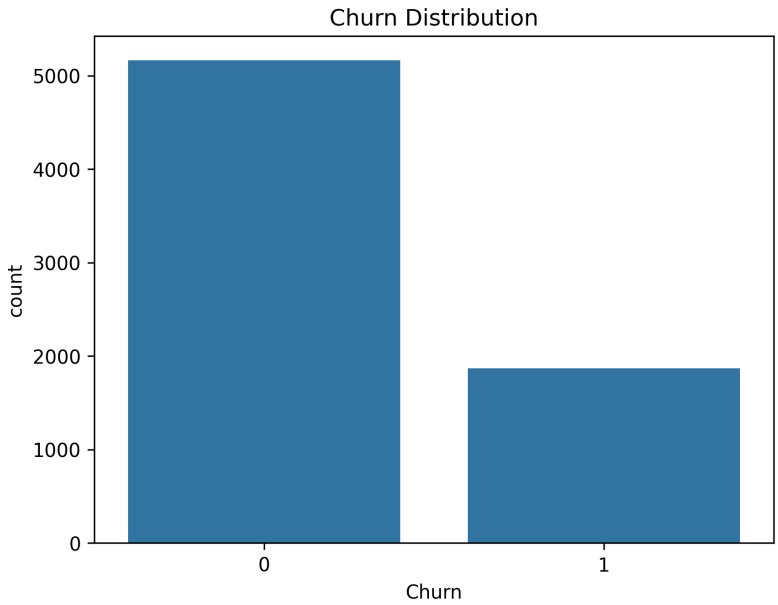

Figure 3.1: Distribution of Churn and Non-Churn Customers

3.3 Train-Test Split

The dataset was divided into:

- 80% Training data (5625 samples)
- 20% Testing data (1407 samples)

3.3.1 Correlation Analysis

The correlation matrix of numerical features is presented in Figure 3.2. It shows the relationship between tenure, monthly charges, total charges, and churn.

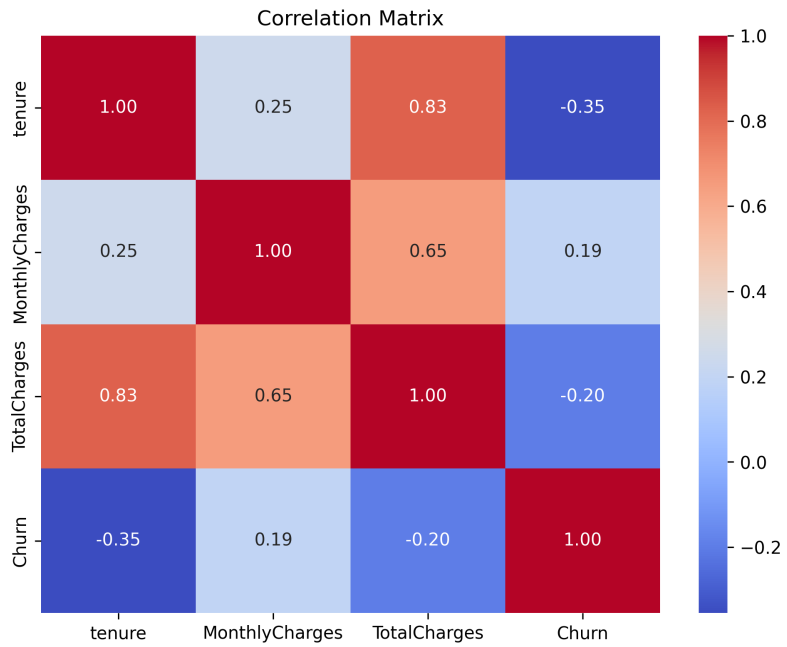

Figure 3.2: Correlation Matrix of Numerical Features

3.4 Models Implemented

3.4.1 Logistic Regression

Logistic Regression was used as a baseline classifier due to its simplicity and interpretability.

3.4.2 Decision Tree

A Decision Tree classifier was implemented to capture non-linear relationships within the dataset.

3.5 Mathematical Formulation of Logistic Regression

Logistic Regression models the probability of customer churn using the sigmoid function.

Let the feature vector be:

$$X = (x_1, x_2, x_3, ..., x_n)$$

The linear model is defined as:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$$

The probability of churn is calculated using the sigmoid function:

$$P(Y = 1|X) = \frac{1}{1 + e^{-z}}$$

Where:

- $Y = 1$ represents churn
- β_i are model coefficients

The model is trained by minimizing the binary cross-entropy loss:

$$L = -\frac{1}{m} \sum_{i=1}^{m} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Chapter 4

Results and Discussion

4.1 Evaluation Metrics

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC score

4.2 Logistic Regression Results

The Logistic Regression model achieved an accuracy of 78.7%. The confusion matrix is shown in Figure 4.1.

- Accuracy: 78.75%
- Precision (Churn): 0.62
- Recall (Churn): 0.52
- F1-score (Churn): 0.56

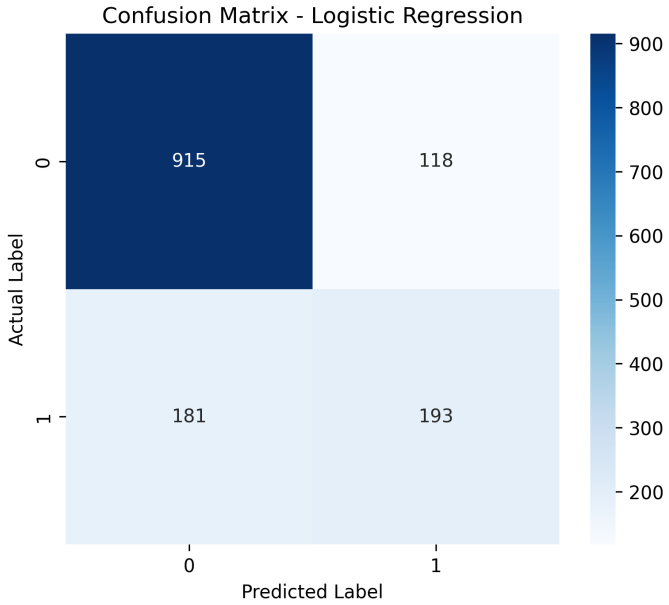

Figure 4.1: Confusion Matrix for Logistic Regression

The relationship between actual churn values and predicted probabilities is shown in Figure 4.2.

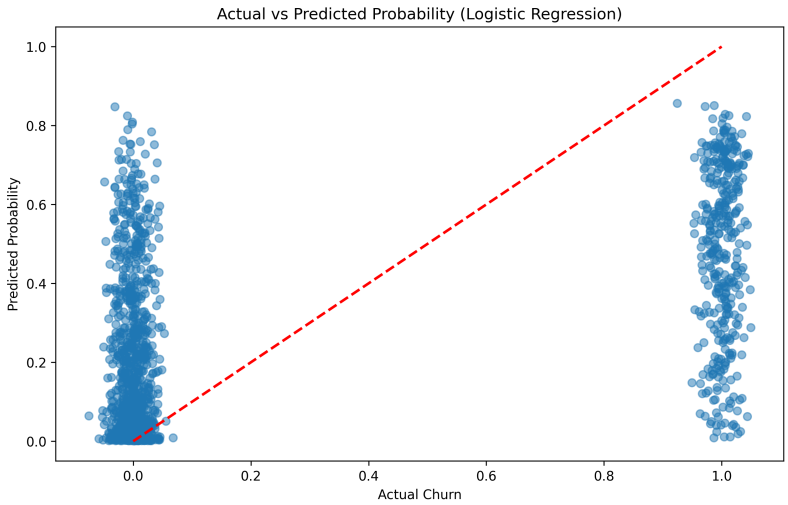

Figure 4.2: Actual vs Predicted Probability for Logistic Regression

4.3 Decision Tree Results

- Accuracy: 72.49%
- F1-score (Churn): 0.50

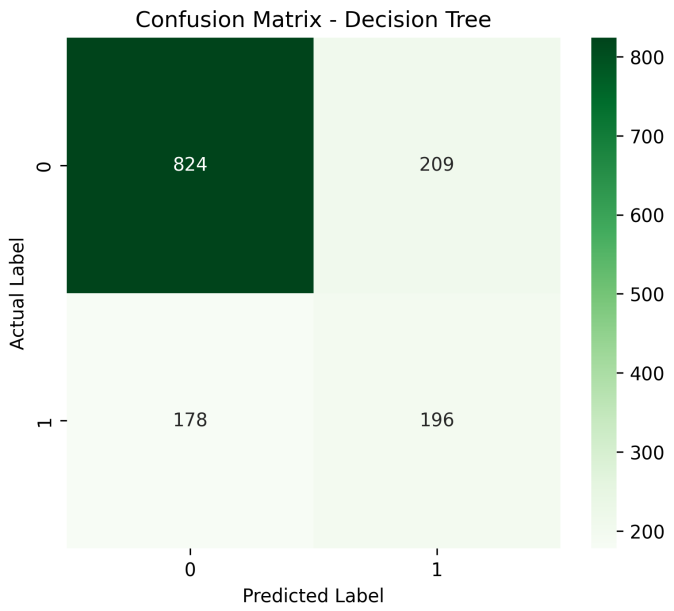

Figure 4.3: Confusion Matrix for Decision Tree

4.4 Handling Class Imbalance using SMOTE

To address class imbalance in the dataset, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data. After applying SMOTE, recall for the minority churn class improved significantly.

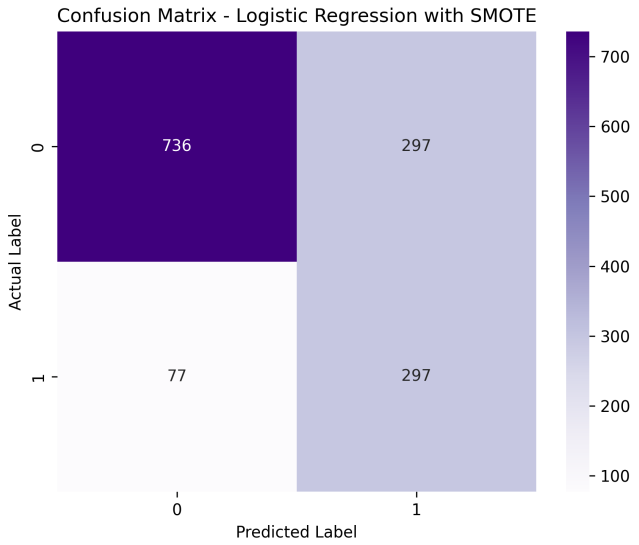

Figure 4.4: Confusion Matrix for Logistic Regression with SMOTE

4.5 Model Comparison

Logistic Regression outperformed Decision Tree in terms of overall accuracy and F1-score. The Logistic Regression model achieved a ROC-AUC score higher than the Decision Tree, indicating better discriminative performance.

Thus, Logistic Regression was selected as the final model.

Model	Accuracy	F1-Score
Logistic Regression	78.75%	0.56
Decision Tree	72.49%	0.50

Table 4.1: Model Performance Comparison

4.6 Evaluation Metrics

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project successfully implemented a machine learning-based customer churn prediction system using the Telco dataset.

Among the evaluated models, Logistic Regression achieved the best performance with an accuracy of 78.75%. The model provides reliable predictions and maintains interpretability, making it suitable for business deployment.

5.2 Future Work

Future improvements may include:

- Implementation of ensemble models such as Random Forest or XGBoost
- Hyperparameter tuning
- Handling class imbalance using SMOTE
- Deployment as a web-based application
- Integration of Generative AI for automated business insights

Appendix A

Appendix

A.1 Appendix 1

A.2 Appendix 2

References

[1] Kaggle, “Telco customer churn dataset,” https://www.kaggle.com/datasets/blastchar/telco-customer-churn, 2018.

[1]